Module 8, Lesson 1

AI 101 for Mechanical Engineers

Classes, Scores and Thresholds

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Lesson Outline

1. Class Scores and Decision Thresholds
2. Confusion Matrix and Classification Outcomes
3. Threshold Selection and Validation

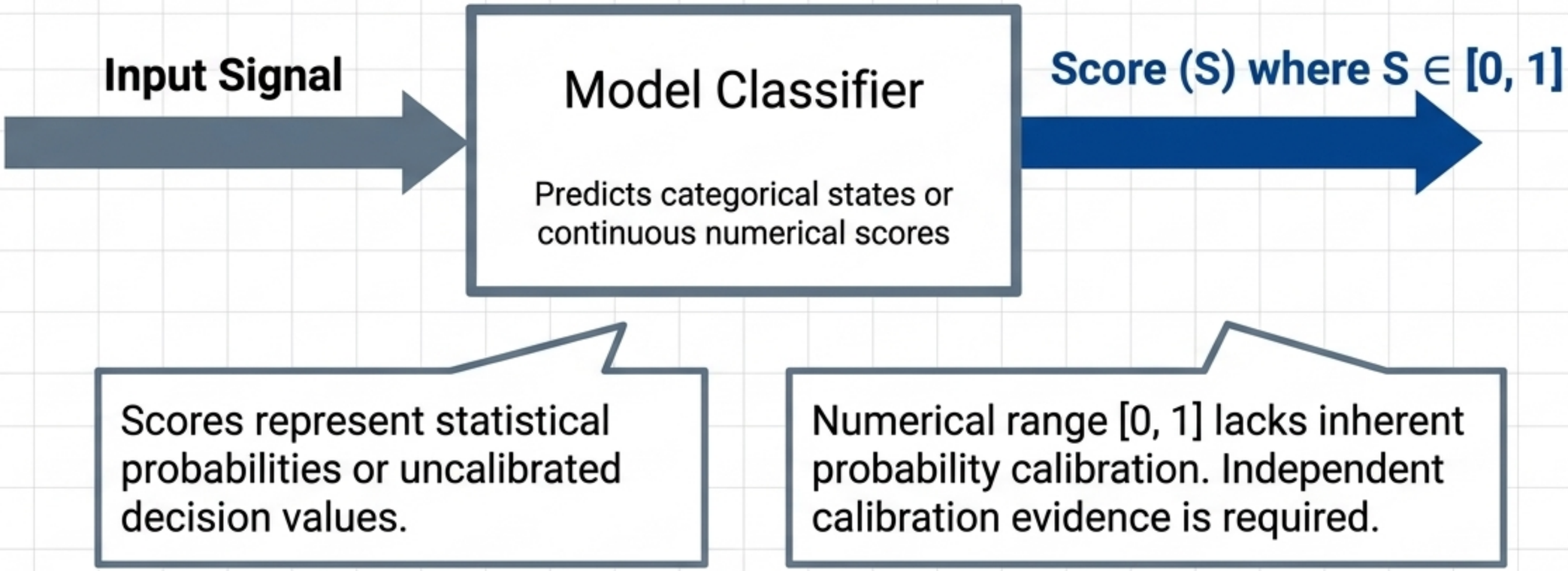
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1. Class Scores and Decision Thresholds

Purpose: Define classifier outputs and formal boundary conditions.

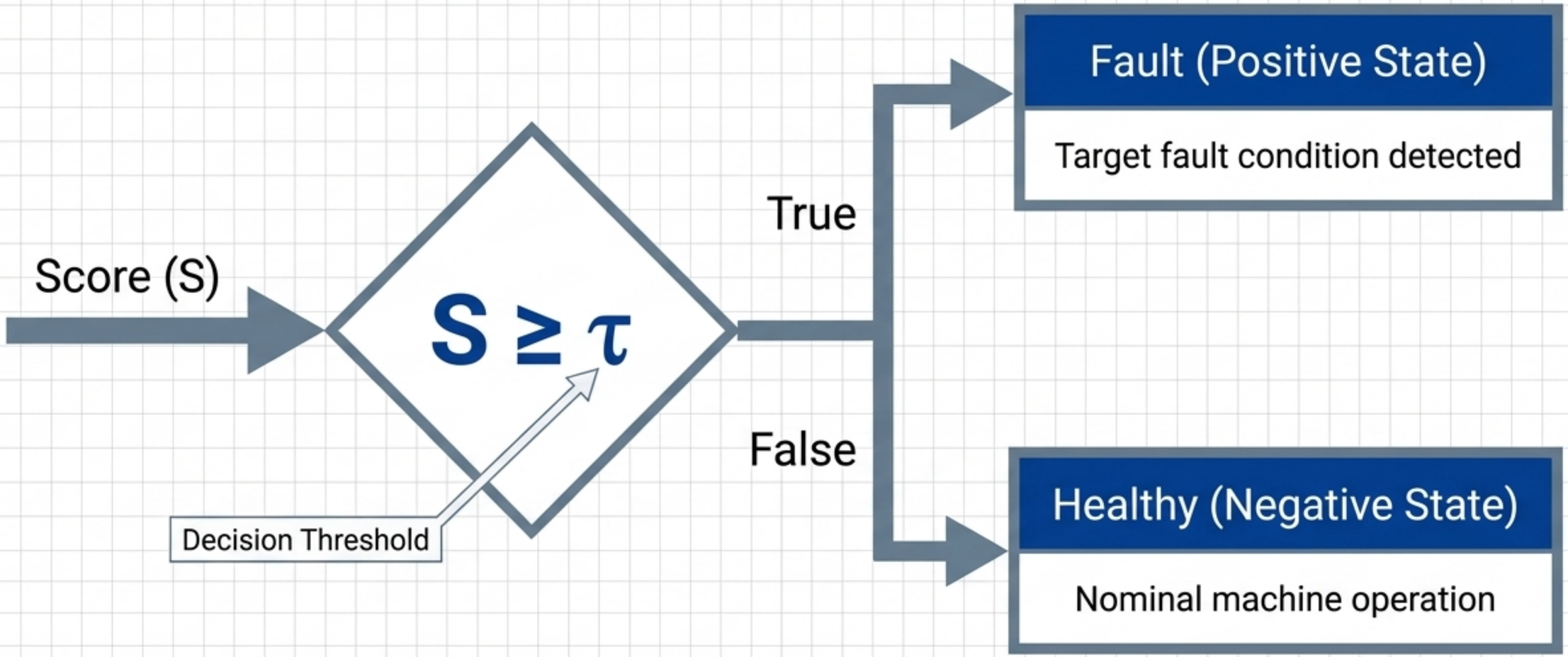
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Classifier Output Formats



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Binary Classification Boundary



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2. Confusion Matrix and Classification Outcomes

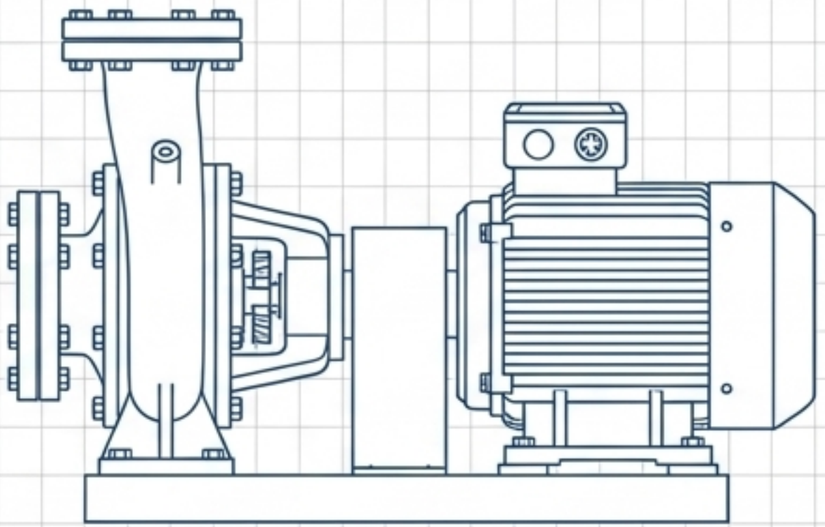
Purpose: Categorize prediction accuracy and error modalities.

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Target Class Specification

Binary classification necessitates a declared positive target class.

System State	Class Designation
Mechanical Fault	Positive [+]
Standard Healthy	Negative [-]



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Prediction Outcome Matrix

Evaluated using a specified dataset and fixed decision threshold. Records aggregate counts against absolute ground truth.

Actual Physical State	Actual Fault	Predicted Healthy
Predicted Fault	True Positive (TP)	False Negative (FN)
Actual Healthy	False Positive (FP)	True Negative (TN)

System Prediction

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Analyzing Classification Errors

FN

Critical Error: Missed Fault

Represents an undetected actual mechanical fault.

FP

Operational Error: False Alarm

Represents an erroneous system fault alarm.

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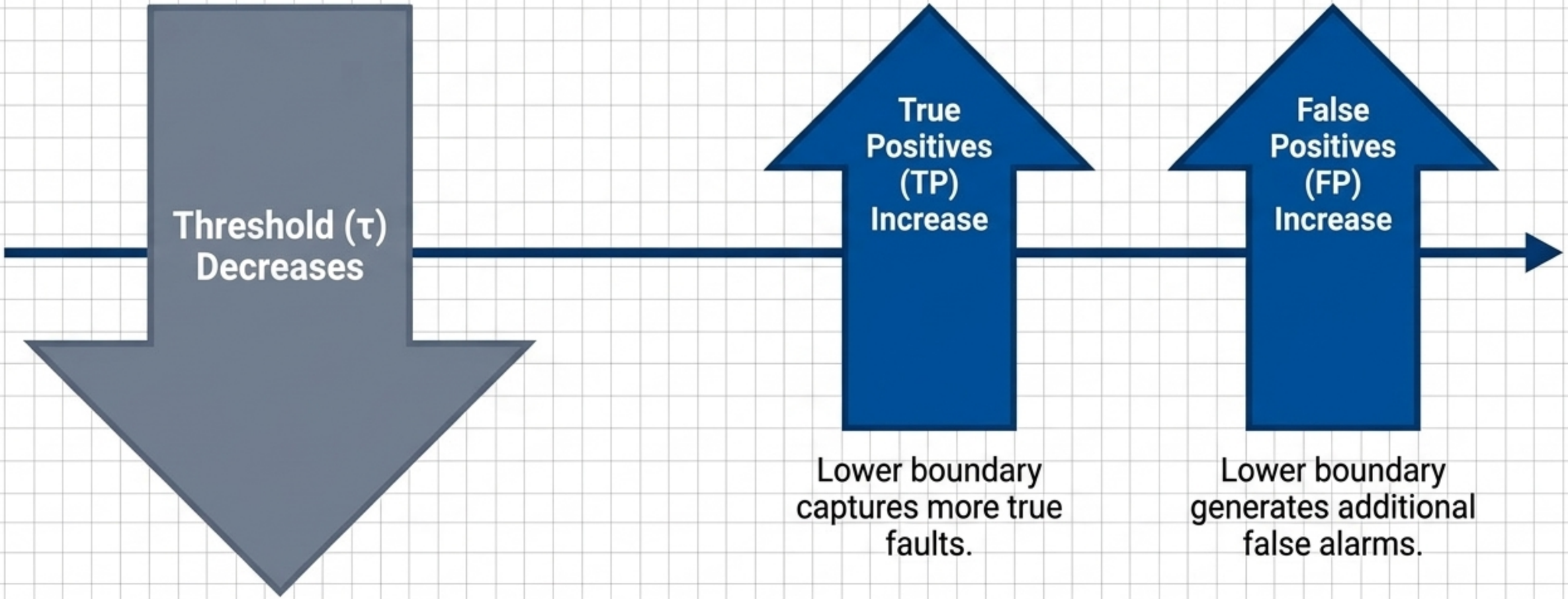
3. Threshold Selection and Validation

Purpose: Optimize decision boundaries using independent dataset evaluation.

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Decision Boundary Trade-offs

Threshold modifications alter final classification distributions.

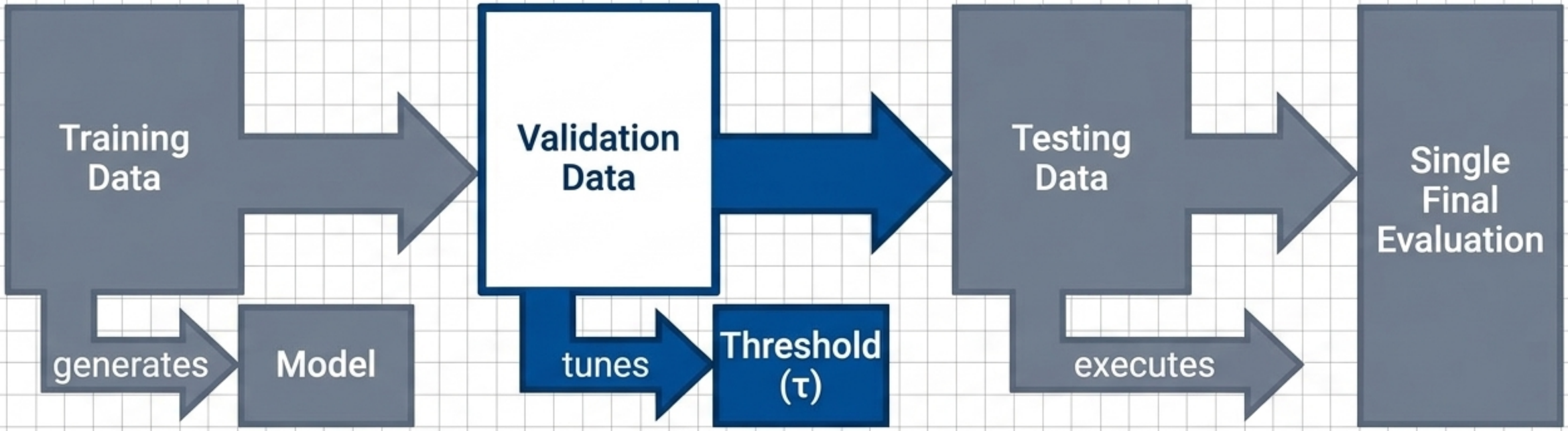


Lower boundary captures more true faults.

Lower boundary generates additional false alarms.

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Independent Threshold Calibration



Balance false alarm costs against missed fault consequences.

Strictly prohibits threshold iteration on final data.

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Key Points

1. Scores require formal thresholds to execute binary fault decisions.

2. Confusion matrices quantify classification performance across distinct modalities.

3. Thresholds demand validation-data tuning to balance mechanical error consequences.

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